

OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

Examination Session and Year	Winter 2020
Module Code	AC2204
Module Title	INTRODUCTION TO MANAGEMENT ACCOUNTING
Paper Number	1
External Examiner	Professor Ivo De Loo
The Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Instructions to Candidates	Answer three questions in total. Please attempt QUESTION ONE from Section A and TWO questions from Section B. Separate Answer books are NOT required.
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings.

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THEN ENSURE THAT YOU HAVE THE CORRECT EXAM PAPER

SECTION A: COMPULSORY

QUESTION ONE

Applet Ltd makes and sells a single product. The following data relate to its first two years of operation.

	Year ended 31/08/2019	Year ended 31/08/2020
Sales (units)	2,000	2,600
Selling Price per unit	€22.50	€23.00
Production (units)	2,300	2,400

Total Costs:		
Variable manufacturing	€24,840	€28,800
Fixed manufacturing	€12,190	€13,200
Variable marketing	€4,000	€5,300
Fixed marketing	€5,000	€5,000

Required:

(A) Calculate, for each of the two years, the unit cost and the operating profit using:

- (i) Marginal costing;
- (ii) Absorption costing.

(25 marks)

(B) Explain why the format of the income statement differs under each method of costing.

(5 marks)

(C) Prepare, for each of two years, a reconciliation which explains the difference in the operating profits resulting from the use of the two costing methods.

(10 marks)

(Total: 40 marks)

SECTION B
ANSWER ANY TWO QUESTIONS

QUESTION TWO

Olaf Ltd manufactures wooden tree houses for children. You have been asked to help plan for the coming financial year. The company currently produces one single tree house model called “Crann” for which the following data is available.

Unit sales price	€1,200
Variable costs per unit	€720
Total fixed expenses	€120,000

Required:

(A) Olaf Ltd plan to sell 300 tree houses (Cranns) in the coming financial year. Calculate each of the following and briefly explain their usefulness for financial planning to the management of Olaf Ltd:

- (i)** Breakeven point in terms of units;
- (ii)** Breakeven point in sales revenue, using the contribution margin ratio;
- (iii)** Operating profit or loss;
- (iv)** Percentage margin of safety;
- (v)** Degree of operating leverage.

(20 marks)

(B) Management at Olaf Ltd are concerned about level of subjectivity there appears to be in estimating the future costs of the organisation. Prepare a short note outlining two possible methods that could be used to address their concerns.

(10 marks)

(Total: 30 marks)

QUESTION THREE

Echo Ltd make three products for which the details shown in the table below are available in relation to the last financial period. A conventional (absorption) costing system is currently in use but an activity-based approach is also being considered by the Board of Management. The Board are not in agreement about this however and at a recent meeting the Director of Manufacturing, Mr John Crowley, stated:

“I don’t know why there is such a fuss about activity-based costing. A cost is a cost and surely it’s enough to know whether it’s a product cost or a period cost.”

<<note: the question is continued on the next page>>

	Total units produced	Material costs per unit	Machine hours per unit	Total machine hours	Labour hours per unit
Product A1	200	48	2	400	4
Product B2	700	23	5	3,500	2
Product D4	300	36	4	1,200	1

Direct labour costs €30 per hour and production overheads are absorbed in the conventional system on a machine hour basis at a rate of €24 per machine hour. Further analysis has indicated that the total production overheads (€122,400) can be divided as shown below.

Costs relating to quality	30%	[Cost driver: Units]
Costs relating to materials handling	25%	[Cost driver: Set-ups]
Costs relating to purchasing	15%	[Cost driver: Purchase orders]
Costs relating to maintenance	30%	[Cost driver: Machine hours]

Activity volumes for the financial period are detailed in the next table.

	Total number of purchase orders	Total number of set-ups
Product A1	10	10
Product B2	6	7
Product D4	4	3
Total	20	20

Required:

- (A)** Calculate the total costs and the cost per unit for each product, using:
- (i)** Traditional (absorption) costing, with production overheads allocated based on machine hours;
 - (ii)** Activity-based costing, with production overheads allocated based on the cost drivers listed in the question.
- [Please show your workings clearly.]

(22 marks)

- (B)** Prepare a short note, commenting on the view expressed by the Director of Manufacturing. In your note, explain how costs are considered in an activity-based costing system.

(8 marks)

(Total: 30 marks)

QUESTION FOUR

Consider each of the following statements. For each statement, state whether you agree or disagree, explaining your answer in each case.

- (A) Decimal Ltd produces a single product. If total variable expenses are €73,000, total fixed costs are €27,930, total sales are 20,000 units, totalling €122,000; then break-even point is 11,400 units, the margin of safety is 43% and operating leverage is 4.6 times.
- (B) Fixed costs per unit remain constant as the level of activity increases within the relevant range.
- (C) Operating leverage at a given level of sales in a capital intensive company will tend to be lower than in a labour intensive company with the same total revenues and total expenses.
- (D) The following monthly activity levels and associated costs were recorded at Moto Ltd for the 2019 financial year.

HIGH: May	3,192 units	Total cost: €69,240
LOW: November	1,963 units	Total cost: €44,660

If it is planned to produce 2,100 units in December 2020, then based on the cost function for 2019, the expected cost is €47,400.

- (E) Opportunity costs are costs that record the difference between two or more decision alternatives.

(5 parts * 6 marks each = Total: 30 marks)